# RAG Assignment: Baseline Retrieval vs. Cross-Encoder Re-Ranking

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code: https://github.com/HRITIK-RANJAN/RAG-BASELINE-VS-RERANKING

## Setup & Environment

- **Dataset:** *Netrunner Comprehensive Rules v26.03*, a 99-page rulebook with many restrictions and exceptions.
- **Models:** `all-MiniLM-L6-v2` embeddings, FAISS cosine search, `ms-marco-MiniLM-L-6-v2` re-ranking, and Gemini Flash Lite for answers (temperature 0).
- **Corpus and machine:** 602 structure-aware chunks were used. Both pipelines use the same frozen corpus.

## 1 Two-Stage Retrieval

### 1.1 Pipeline 1: baseline retrieval

The bi-encoder embeds the query and all chunks independently. FAISS returns the top 10 by cosine similarity, and the top 3 are sent to the LLM.

### 1.2 Pipeline 2: advanced re-ranking

The second pipeline re-scores those same 10 candidates with the Cross-Encoder, then sends its top 3 to the identical LLM prompt.

## 2 Architecture comparison

A **Bi-Encoder** creates independent embeddings for the query and each chunk, then compares them with cosine similarity. Chunk vectors can be computed offline, making it fast, but word relationships such as "cannot" or "only if" may be blurred.

A **Cross-Encoder** reads the query and chunk together with joint self-attention across their tokens and gives one pairwise score. This captures conditions and wording better, but requires a separate model pass for every candidate at query time.

## 3 Failure Mode Analysis (Stress Testing)

**Query set.** The final set contains 23 stress-test queries covering controls, keyword traps, exceptions, negation and buried answers. It is not a random sample.

| | Hit@3 | Recall@3 | MRR@10 | Mean gold rank (in top 10) |
|---|---|---|---|---|
| Pipeline 1 (baseline) | 0.652 | 0.514 | 0.637 | 2.29 |
| Pipeline 2 (re-ranked) | **0.696** | **0.638** | **0.687** | **2.24** |

Table 1: Retrieval metrics on the 23 final queries (essential gold chunks only).

By category (Hit@3 baseline → re-ranked): buried 0.40 → 0.60, control 0.83 → 1.00, exception 0.67 → 0.67, keyword trap 0.60 → 0.60, negation 0.75 → 0.50. Re-ranking moved the best gold chunk up in 5 queries, down in 6 and left it in place in 12. The two models agree only moderately on the order of the same 10 chunks (median Spearman 0.38, range −0.20 to 0.94).

**Examples.** Q29 is a keyword trap: "flip face up" refers to the rule term "rez". Its relevant chunk moved from rank 4 to rank 1. Q02 and two buried-answer queries also moved into the top 3.

**Limitations.** Re-ranking also hurt: Q36 moved from rank 1 to rank 9 and Q23 from rank 1 to rank 7. Q12 was outside the initial top 10, so no re-ranker could recover it.

### 3.1 Answer 1 vs. Answer 2

Answer 1 had 15 correct answers, 7 refusals and 1 hallucination. Answer 2 had 16 correct answers, 6 refusals and 1 hallucination. Four baseline refusals became correct after the relevant chunk entered the top 3; three correct

| ID | Category | Gold chunk: bi → CE rank | Query |
|---|---|---|---|
| *Fixed by re-ranking* | | | |
| Q02 | control | 5.2.6: #7 → #1 | What basic actions can the Corp always take during their turn? |
| Q29 | keyword trap | 8.1.2e: #4 → #1 | How does the Corp flip an installed card face up so that it starts working? |
| Q35 | buried | 7.4.7: #4 → #1 | In what order are cards from the Corp's deck offered to the Runner when R&D is breached? |
| Q41 | buried | 7.3.8: #4 → #1 | If stealing an agenda would cause a breach of the same server during an ongoing breach, when does the second breach happen? |
| *Hurt by re-ranking* | | | |
| Q23 | negation | 3.8.5: #1 → #7 | Can the Runner have two consoles installed at the same time? |
| Q32 | keyword trap | 4.2.7: #3 → #4 | What happens when the Corp runs out of cards in their deck and has to draw? |
| Q36 | buried | 7.5: #1 → #9 | What does the Runner do in a breach when there are no more cards left to choose from? |
| *Failing in both / not in the top 10* | | | |
| Q18 | negation | 1.2.2: #6 → #6 | If one effect says a player must do something and another says they cannot, which applies? |
| Q34 | buried | 7.4.6: #4 → #5 | While the Runner is breaching HQ, a new card enters the Corp's hand. Can the Runner access it? |
| Q12 | exception | 1.13.13: not in top 10 | What happens to cards hosted on a card that moves from the Corp's to the Runner's score area? |
| Q42 | keyword trap | 10.1.1: not in top 10 | What happens if I have two copies of a card with the diamond symbol in play? |

Table 2: Rank shifts of the gold chunk for every final query whose outcome is not "both ok".

baseline answers became refusals when re-ranking moved that chunk out. Q34 is the clearest failure: the critical chunk was outside the top 3, so both answers hallucinated a "yes" instead of refusing. Thus, re-ranking improved one answer overall, but it did not remove every hallucination or guarantee that an essential clause would be present.

# 4 Latency

Timings exclude model loading and use 20 repeats over the 23 queries. The LLM was timed separately because API latency is variable.

| | Pipeline 1 (ms) | | | Pipeline 2 (ms) | | |
|---|---|---|---|---|---|---|
| Stage | median | mean | p95 | median | mean | p95 |
| Query encoding (bi-encoder) | 4.12 | 4.23 | 4.72 | 4.12 | 4.23 | 4.72 |
| FAISS search, top 10 | 0.02 | 0.02 | 0.03 | 0.02 | 0.02 | 0.03 |
| Cross-encoder, 10 pairs | – | – | – | 35.04 | 34.64 | 36.14 |
| **Retrieval subtotal** | **4.15** | 4.25 | 4.75 | **39.22** | 38.89 | 40.61 |
| LLM generation (Gemini) | 1322.5 | 5036.4 | 7439.2 | 1764.6 | 5418.8 | 14826.3 |
| End-to-end | 1326.6 | 5040.6 | 7443.3 | 1802.3 | 5457.6 | 14865.3 |

Table 3: Latency per query (Apple M4, MPS).

Re-ranking adds about 35 ms per query. This is much slower than baseline retrieval, but small beside LLM generation, which dominates end-to-end time.

# 5 Why not cross-encode the whole corpus?

The Cross-Encoder cannot precompute a score because it needs the query and chunk together. Its cost therefore grows linearly with the number of chunks: about **2.74 ms per pair**. The Bi-Encoder first narrows the collection cheaply, leaving only 10 pairs for accurate re-ranking.

| Chunks $N$ | Cross-Encoder on all $N$ | Two-stage (encode + exact search + CE on 10) |
|---|---|---|
| 602 | 1.65 s | 0.04 s |
| 100,000 | 274 s (4.6 min) | 0.04 s |
| 1,000,000 | 2,736 s (46 min) | 0.07 s |
| 10,000,000 | 27,364 s (7.6 h) | 0.37 s |

Table 4: Time per query (above 200k chunks: linear extrapolation; approximate indexes would make the two-stage column grow even more slowly).




Figure 1: Rank of the 10 candidates before (left) and after (right) re-ranking for every non-"both ok" query. Red = essential gold chunk, green band = the top 3 sent to the LLM.

For this corpus, re-ranking all 602 chunks would take about 1.66 s, compared with about 0.04 s for the two-stage retrieval. At 1 million chunks, the same approach would take roughly 46 minutes per query. The trade-off is a recall ceiling: a relevant chunk missed by the first stage cannot be recovered.

# Limitations

The query set was selected for stress testing, and timings are from one machine. LLM latency is noisy; scaling figures beyond the measured corpus are extrapolations.